GNN-Based Proactive Congestion Pre-emption — Related Work

Project summary

We predict congestion on data center switch links milliseconds before it happens and reroute traffic early. A spatio-temporal graph neural network (GNN) reads live in-band network telemetry (INT) from every switch and forecasts per-link queue build-up across the whole fabric. Existing systems react only after queues grow or packets drop.

Top 5 related papers

Ranked by overlap with our idea. NTP-INT is the closest and should be read first.

#	Paper	Venue / year	What it does	Key difference from ours
1	NTP-INT: Network Traffic Prediction-Driven In-band Network Telemetry for High-load Switches	arXiv, 2025	Multi-Temporal GNN (MTGNN) forecasts traffic to find high-load switches; deep RL plans INT probe paths; cuts control overhead by 50%	Prediction decides where to collect telemetry, not how to reroute traffic
2	GraphCC: A Practical Graph Learning-based Approach to Congestion Control in Datacenters	arXiv, 2023	Multi-agent RL + GNN agents on switches cooperate to tune the global ECN configuration	Tunes ECN marking thresholds; no congestion forecasting or rerouting
3	RouteNet-Fermi: Network Modeling with Graph Neural Networks	IEEE/ACM ToN, 2023	GNN predicts delay, jitter and loss; scales to 300-node networks; 6.24% mean relative error on delay. Builds on RouteNet	Offline planning and what-if model; not live, millisecond-scale prediction from INT

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4	HPCC: High Precision Congestion Control	ACM SIGCOMM, 2019 (Alibaba)	Uses INT link-load data for precise sender rate control; up to 95% shorter flow completion times vs DCQCN and TIMELY	Reactive, per-flow at the sender; no ML, no fabric-wide prediction. Our main baseline
5	SMOOTHIE: INT + SRv6 dynamic path load balancer	Université de Liège	P4 switches collect INT; SRv6 reroutes flows; beats ECMP, competitive with CLOVE	Threshold-based rerouting; no GNN forecasting

Novelty risks

Two works sit close enough that our claim must clearly differ from them.

- RouteNet-TGNN: A Temporal Graph Neural Network (Journal of Communications, 2026). A temporal extension of RouteNet aimed at proactive SDN decisions, including congestion mitigation. This is the biggest risk; read it in full.
- US Patent 10,938,722 — In-band telemetry congestion control system. Detects and controls congestion from INT data, such as queue occupancy, carried by P4 switches. Existing patent prior art for INT-based congestion handling.